A Reversible Rational McMillan Map: Exact Pole Exclusion and Low-Period Monodromy

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Abstract

We examine the rational recurrence $M(x,y) = (-4x/(1+x^2) - y, x)$. Exact symbolic calculation verifies its rational inverse, coordinate-swap reversor, unit Jacobian determinant, and quartic first integral. The valid fixed locus over $\mathbb{C}$ contains one real and two nonreal points, while $(1,-1) \leftrightarrow (-1,1)$ is a genuine real primitive two-cycle. We expose and remove the factor $(x^2+1)^2$ introduced by denominator clearing: its roots are poles, not periodic points. The cycle's two-step monodromy is $-I_2$. These are finite low-period certificates, not a global orbit classification or transfer determinant.

1 Rational map and invariant

We freeze the McMillan/QRT-type family

$$M_\mu(x,y) = \left(\frac{2\mu x}{1+x^2} - y, x\right), \qquad \mu = -2.$$

Over $\mathbb{C}$ the forward map excludes $x^2+1=0$. Its rational inverse is

$$M^{-1}(x,y) = \left(y, -\frac{4y}{1+y^2} - x\right).$$

For the involution $S(x,y) = (y,x)$, direct composition gives $SMS = M^{-1}$. Differentiation gives

$$DM(x,y) = \begin{pmatrix} 4(x^2-1)/(1+x^2)^2 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \det DM = 1.$$

Substitution and cancellation also give

$$I \circ M = I, \qquad I(x,y) = x^2y^2 + x^2 + y^2 + 4xy.$$

2 Fixed points and a two-cycle

The fixed equations force $y = x$ and reduce to

$$-\frac{2x(x^2+3)}{x^2+1} = 0.$$

Thus the valid fixed points are $(0,0)$ and $(\pm i\sqrt{3}, \pm i\sqrt{3})$, with matched signs. Only the origin is real. Direct substitution gives the distinct real cycle

$$q_+ = (1,-1) \longmapsto q_- = (-1,1) \longmapsto q_+, \qquad I(q_\pm) = -1.$$

Both cycle denominators equal 2, so neither point is a pole.

For the second iterate, eliminating y from the two cleared numerators gives

$$-8x(x-1)(x+1)(x^2+1)^2(x^2+3).$$

The roots of x^2+1 cannot be tested as orbit points because the first application of M is already undefined. Removing this pole factor leaves $x(x-1)(x+1)(x^2+3)$: the factors $x(x^2+3)$ recover the fixed locus and x^2-1 gives the primitive cycle. The domain audit is therefore

candidate	type	$1+x^2$	status
0	real fixed	1	valid
$\pm i\sqrt{3}$	nonreal fixed	-2	valid
± 1	real two-cycle	2	valid
$\pm i$	cleared-denominator root	0	excluded

3 Local period-two monodromy

At either cycle point the first derivative is

$$J_\pm = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

In chronological order the derivative of M^2 is therefore

$$P_2 = J_- J_+ = -I_2, \qquad \det P_2 = 1, \quad \operatorname{tr} P_2 = -2.$$

Consequently

$$\det(\lambda I - P_2) = (\lambda + 1)^2, \qquad \det(I - zP_2) = (1+z)^2.$$

Proposition 1. *The inverse, reversor, invariant, fixed points, primitive cycle, and displayed monodromy identities are exact on their stated domains.*

Proof. An independent checker recomputes both rational compositions, differentiates the map, substitutes into I, verifies every orbit arrow, and multiplies the Jacobians without importing the evidence producer. □

4 Route-A boundary

The polynomial $(1+z)^2$ belongs only to the local derivative of M^2 along one cycle. No transfer operator, finite transfer owner, function space, or tail bound has been constructed. The result is `A1_PARTIAL_CERTIFIED`, `A2_FAIL`, `A3_NOT_ADDRESSED`, and `A4_FAIL`. We do not claim a complete orbit atlas, global integrability classification, analytic Fredholm determinant, arithmetic data, or Route-B object. The scope firewall is `NO_BAD_EULER_OR_ROOT_NUMBER`.